Program 4:
Integrate Git with Docker for source-controlled applictaion build
07/11/2025

Project Directory:

prog4
- |- Dockerfile
- |- app.py
- |- requirements.txt

Program code and steps for execution:

Step 1: Create a folder with the name prog4 for the program 4 execution
> mkdir prog4
> cd prog4/

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog4
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog4/
```

Step 2: Create the three files
1. app.py 2. Dockerfile 3. requirements.txt
And the below images shows the content to be included in each file

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ touch app.py requirements.txt Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ ls
app.py  Dockerfile  requirements.txt
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ nano app.py
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ cat app.py
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Program 4: Flask App running inside Docker!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

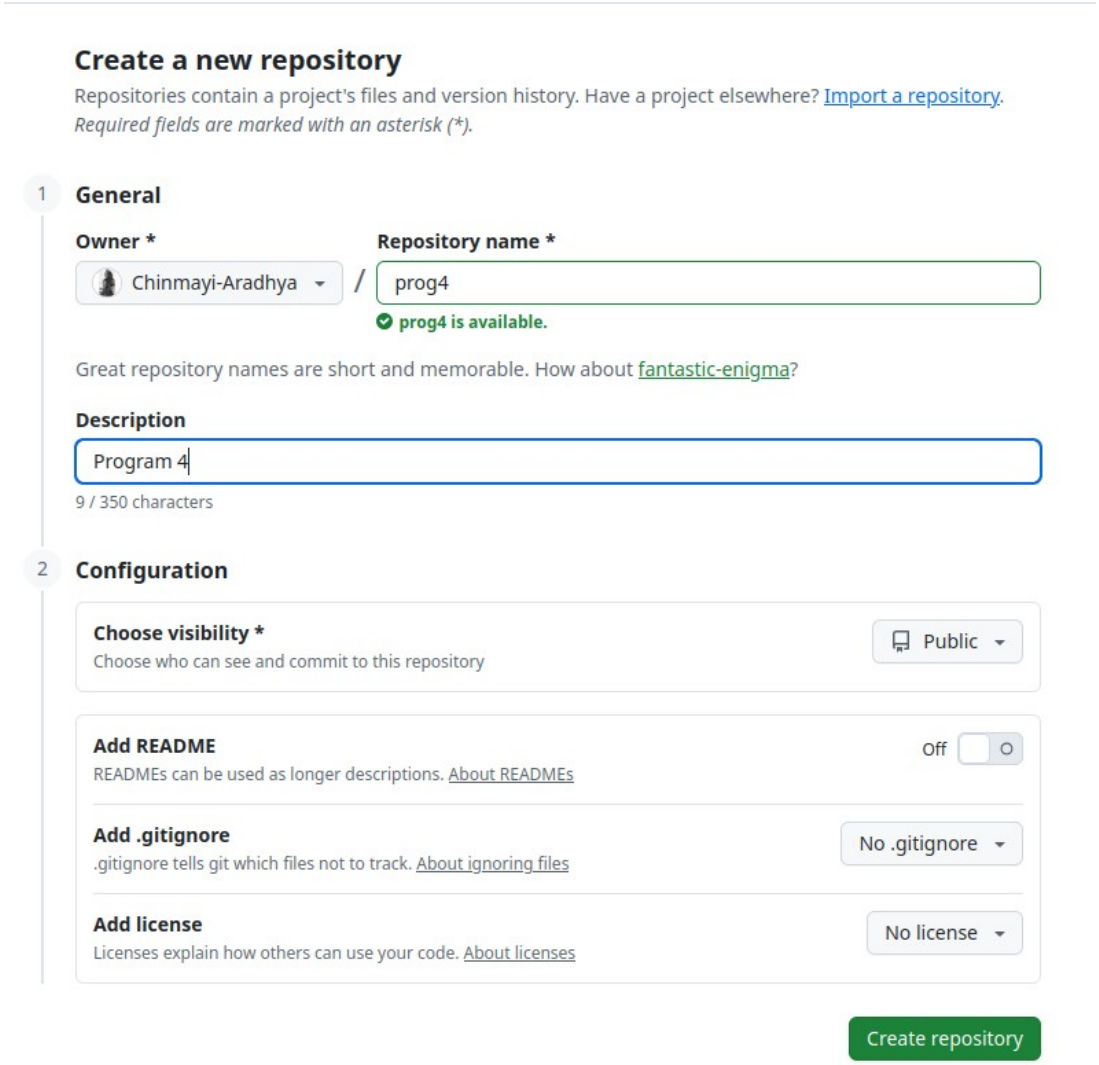
```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ nano requirements.txt
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ cat requirements.txt
Flask==2.2.5
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ nano Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ cat Dockerfile
FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

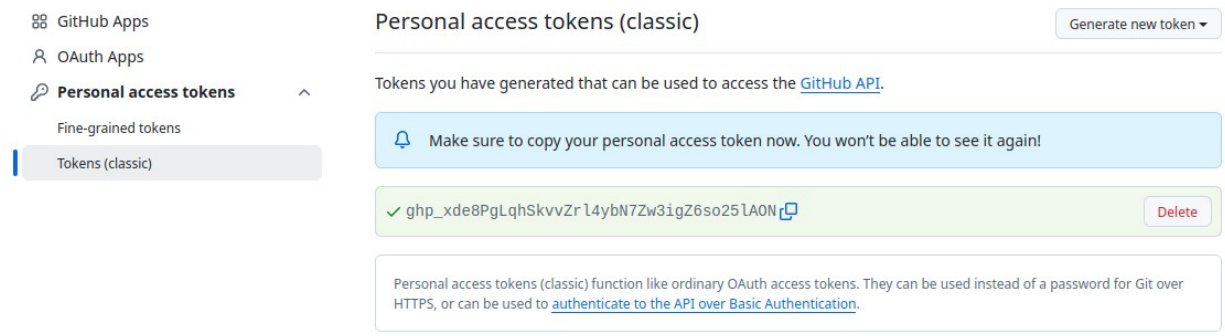
Step 3: set the Git name and email first, then initialize
> git config --global user.name "<username>"
> git config --global user.email "<user-email>"
> git init
> git add .
> git commit -m "Initial commit for Program 4"

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ git config --global user.name "Chinmayi-Aradhya"
git config --global user.email "chinmayiaradhya03@gmail.com"
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog4$ git init
git add .
git commit -m "Initial commit for Program 4"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/chinnu/devops_lab/prog4/.git/
[master (root-commit) f6f4fdf] Initial commit for Program 4
 3 files changed, 22 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
```

Step 4: Login to your github account from browser and create a new repository with the name prog4



Step 5: In the github page follow the below order
> login > settings > developer settings > Personal Access Tokens > Token (classic) > Generate new token > copy the token



Step 6: Push the project to GitHub using the following commands,
> git remote add origin https://github.com/<your-username>/prog4.git
> git branch -M main
> git push -u origin main
Here after the above commands it will ask the user name and password then paste the token which we have created

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/p
rog4$ git remote add origin https://github.com/Chinmayi-Aradhya/prog4.git
git branch -M main
git push -u origin main
Username for 'https://github.com': Chinmayi-Aradhya
Password for 'https://Chinmayi-Aradhya@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 621 bytes | 621.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Chinmayi-Aradhya/prog4.git
```